

Lattice Boltzmann description of magnetization in porous media

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The magnetic moments of fluid particles filling the pore space of a porous material (1) reside in a complex space, (2) are carried by the particles in diffusive exploration of the pore space, and (3) relax when the particles approach relaxation sites on the walls of the pore space. Further, when the magnetic moments are the object of a nuclear magnetic resonance experiment, they are manipulated by rf magnetic fields, internal magnetic field gradients, and applied magnetic field gradient pulses. In this paper, a lattice Boltzmann computational procedure is described that accounts for all of the vagaries in the experience of a magnetic moment in a porous material. The time evolution of the longitudinal and transverse magnetization, in a variety of experimental situations, can be simulated with this computational procedure. The z component of the magnetization, the longitudinal magnetization, is described by a set of coarse grained distribution functions for a scalar fluid. The time evolution of these distribution functions involves a scattering process (to account for diffusion) and a probability of transmission out of the pore space (to account for surface relaxation). A numerical example, involving a pore adjacent to a microporous region, is examined in detail. The transverse magnetization is a vector. Its x and y magnetization components are carried by separate scalar fluids. There is a set of coarse grained distribution functions for each fluid. Radio frequency magnetic fields, internal magnetic field gradients, applied magnetic field gradient pulses, etc., represent conversion processes in which the two fluids transform into one another. Two examples, one involving a periodic field gradient and a Hahn echo, and the other involving an isolated pore and a PFG sequence, are examined in detail.

I. INTRODUCTION

The time evolution of the magnetization of a fluid $M(t)$ in a pore space is in many instances controlled by the geometry of the pore space. Thus nuclear magnetic resonance (NMR) measurements are one of the principle ways to learn about pore space geometry.^{1,2} Sophisticated NMR tools,¹ and elaborate experimental protocols^{3,4} are an essential part of current practice and development.

The description of $M(t)$ of a fluid in a pore space has four main ingredients. (1) The fluid resides in a complicated pore space. (2) The magnetic moments carried by the fluid molecules diffusively explore this pore space. (3) The magnetic moments relax when they are carried close to relaxation sites on the walls of the pore space. (4) The time evolution of the magnetization is measured using NMR fields that label, manipulate, and interrogate the magnetic moments. NMR fields include radio frequency (rf) pulses, gradients, and applied gradient pulses. In some instances these fields simply set the initial conditions, as in the case of a T_1 measurement. In other instances these fields modify the magnetic system importantly as the time evolution is occurring.

Given $M(t)$, a magnetization relaxation curve, what is the pore space geometry? This inverse problem is solved by forward calculations using a sequence of models for the behavior of the magnetization in a pore space. The success of solution depends critically on the quality and practicality of the models. Perhaps the best known model of magnetization relaxation in a pore space is the isolated pore model of Brownstein and Tarr.⁵ This model is solved for simple ge-

ometries in terms of the relaxation modes of the magnetization. The solution to this model in the fast diffusion limit is used to solve the inverse problem in which a spectrum of relaxation times is mapped onto a spectrum of pore radii.⁶ The isolated pore model is often used to assess the importance and interplay of physical variables.^{7,8} Models of magnetization relaxation in more complex geometries have been discussed in a number of approximations, e.g., taking the pore space to be periodic,⁹ or assuming the behavior of the magnetization in each pore is described by a single relaxation mode in order to study pore-pore coupling.^{8,10,11} These models provide an understanding of average properties and limiting behavior, but are relatively inflexible. Models of magnetization relaxation in complex geometries have been solved exactly using random walk methods.^{12,13} **Random walk methods are used to confirm approximate solutions and to provide insight in complicated situations. They are too computationally intensive to be used for data analysis.**

Lattice Boltzmann methods have been very useful in "solving" a wide variety of difficult transport problems.^{14,15} **This is because one is able to give up details about the short range behavior of a fluid in return for having complex interactions in the fluid or for having the fluid in a complex space.** The problem at hand is ideally suited to this trade off. The purpose of this paper is to describe in detail the use of lattice Boltzmann methods for solution to problems involving magnetization motion in a pore space. One of the nice things about the lattice Boltzmann computational procedure in this context is that it lends itself naturally to description of the labeling, manipulation, and interrogation of the fluid that are

the special province of NMR investigations.

Section II contains a lattice Boltzmann description of the time evolution of the z component of the magnetization. The computational rules for specifying the pore space geometry, the particle diffusion, and the magnetization relaxation at the surface of the pore space are described. In Sec. III the resulting lattice Boltzmann computational procedure is applied to the relaxation of the magnetization in a pore adjacent to a microporous region of space. In Sec. IV the procedure for handling the transverse magnetization is described. The transverse magnetization is described by two scalar fluids and a conversion process between the two fluids. One scalar fluid carries the x component of the magnetization, and a second scalar fluid carries the y component of the magnetization. Each of these fluids obeys the rules developed in Sec. II. The conversion process between the two fluids accounts for precession of the transverse magnetization in local magnetic fields. The resulting lattice Boltzmann computational procedure for the motion of transverse magnetization is applied to (1) a Hahn echo for magnetization in a sinusoidal gradient and (2) a pulsed field gradient (PFG) probe of the magnetization in a bounded pore space. Section V contains our conclusions and suggestions for future work. Certain useful details are found in two Appendixes.

II. LONGITUDINAL MAGNETIZATION

The time evolution of the z component of the magnetization in a porous material is governed by the equation of motion^{10,11}

$$\frac{\partial M_z}{\partial t} = D \nabla \left[\phi \nabla \left(\frac{M_z}{\phi} \right) \right] - \sum_{i=1}^N \rho_i \delta(\mathbf{x} - \mathbf{x}_i) (M_z - M_\infty) - \frac{M_z}{T_{1B}}, \quad (1)$$

where $\phi(\mathbf{x}) = 1$ in the pore space, $\phi(\mathbf{x}) = 0$ in the matrix, ρ_i is the strength of a relaxation process at site $\mathbf{x}_i$ on the surface of the pore space, and D is the diffusion constant. Assuming the NMR experiment is tracking hydrogen magnetic moments $D = D_{\text{H}_2\text{O}}$, the diffusion coefficient of water, the relaxation rate of the magnetization in bulk fluid T_{1B} is about 3 sec. As T_{1B} is long, the last term in Eq. (1) will be neglected in the simulations. In an NMR experiment the magnetization is created by an external dc magnetic field that polarizes the magnetic moments, thus $M_z(\mathbf{x}, 0) = N\mu$, where N is the number of magnetic moments in the system, and μ is the magnitude of the magnetic moment. The lattice Boltzmann simulation will track $M_z(\mathbf{x}, t)$, given an evolution governed by Eq. (1). We formulate the lattice Boltzmann description of magnetization evolution and show examples of its use in two dimensions. Transcription to three dimensions is straightforward.¹⁴

The lattice Boltzmann distribution functions $F_\alpha(\mathbf{R}, t)$ used to describe a scalar fluid are discrete in time, space, and velocity. Time is discretized in steps of size τ , thus $t_{m+1} = t_m + \tau$. Space is discretized to a hexagonal lattice as illustrated in Fig. 1. Average properties within each hexagon are assigned to its center point. Thus the simulation takes place on a triangular lattice with lattice sites, $\mathbf{R}_{kl} = (l-1)\mathbf{a} + (k-1)\mathbf{b}$, where $l = 1, \dots, L$, $k = 1, \dots, K$, $\mathbf{a} = a(1, 0)$, $\mathbf{b} = a(-1/2, \sqrt{3}/2)$ in an x - y coordinate system, and a is the unit

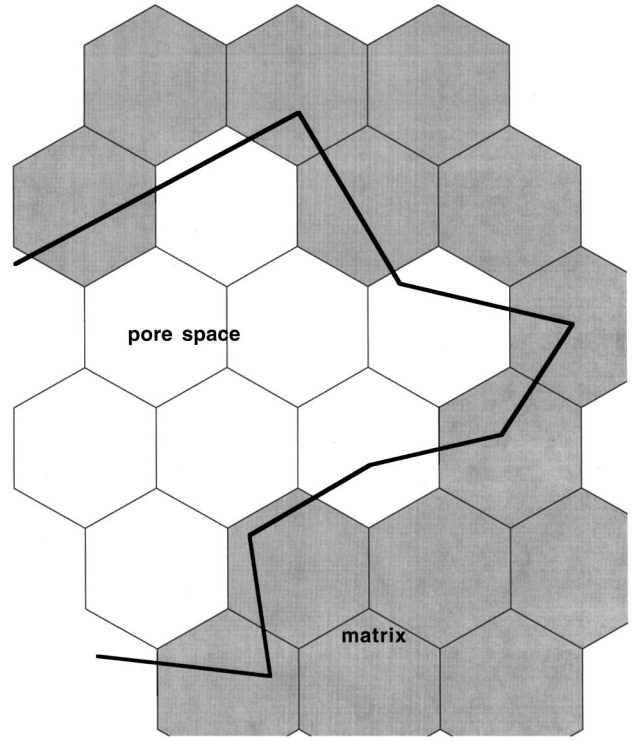


FIG. 1. Discrete space. A two-dimensional porous material is tiled with hexagons. The tiles are located by their centers, creating a triangular lattice of sites at positions $\mathbf{R}$. The heavy line represents the pore/matrix interface. If more than half the area of a hexagon is in the pore space, $\phi(\mathbf{R}) = 1$, otherwise $\phi(\mathbf{R}) = 0$.

of length. Velocity is discretized to six vectors, $\mathbf{v}_\alpha = c\mathbf{e}_\alpha$, where c is the unit of velocity, $\mathbf{e}$ is a unit vector, and $\alpha = 1, \dots, 6$, denoting the six directions from a site $\mathbf{R}$ to its six nearest neighbor sites (see Fig. 2). Time, space, and the magnitude of the velocity are related by $a = c\tau$. In the numerical work, $a = \tau = c = 1$. The distribution functions $F_\alpha(\mathbf{R}, t)$ specify the number of particles at site $\mathbf{R}$, with velocity $c\mathbf{e}_\alpha$, at time t . There are six distribution functions at each site describing the behavior of the particles in the system.

The magnetization in the system is carried by the proton moments of the hydrogen atoms. These moments do not influence the dynamics of the particles. If at time $t = 0$ the magnetic moments are polarized, then $F_\alpha(\mathbf{R}, 0) = \phi(\mathbf{R})$. The

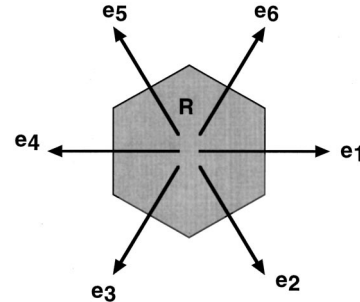


FIG. 2. Discrete velocity. The particle velocity at a site is specified in the 6 directions to nearest neighbor sites. The first unit vector $\mathbf{e}_1$ is in the $+x$ direction. The unit vectors are periodic, thus $\mathbf{e}_{\alpha+1}$ is well defined for $\alpha = 1, \dots, 6$.

initial value of each distribution function is the same for all $\mathbf{R}$ and α since the initial polarization of the magnetic moments is uniform throughout the pore space. The magnitude of the initial value can be chosen for convenience since $M_z(t)$ is measured relative to $M_z(0)$. As the magnetic moment μ is a multiplier of the particle density, we scale it out and study the particle density. Thus the magnetization at $\mathbf{R}$ is

$$M_z(\mathbf{R}, t) = \sum_{\alpha} F_{\alpha}(\mathbf{R}, t) \quad (2)$$

and the total magnetization in the system is

$$M_z(t) = \sum_{\mathbf{R}} \sum_{\alpha} F_{\alpha}(\mathbf{R}, t). \quad (3)$$

The relaxation processes that degrade the magnetization by depolarizing magnetic moments are equivalent to processes that remove particles carrying magnetic moments from the system of particles. Thus by following the time evolution of a system of particles that can appear and disappear under suitable conditions, we follow the time evolution of the magnetization.

A. Computational rules

The geometry of the pore space and the two dynamical processes in Eq. (1), diffusion and relaxation, are incorporated into the rules for the time evolution of $F_{\alpha}(\mathbf{R}, t)$ as follows.

(1) *Geometry: G.* The geometry of the pore space is defined by the state of a site. If site $\mathbf{R}$ is in the pore space it is in state $\phi(\mathbf{R})=1$. If site $\mathbf{R}$ is a matrix site it is in state $\phi(\mathbf{R})=0$. Distribution functions $F_{\alpha}(\mathbf{R}, t)=0$ at sites with $\phi(\mathbf{R})=0$.

(2) *Translation: T.* In the absence of diffusion and relaxation processes, the distribution functions simply translate in one time step

$$F_{\alpha}(\mathbf{R}, t_{m+1}) = F_{\alpha}(\mathbf{R} + a\mathbf{e}_{\alpha+3}, t_m) \phi(\mathbf{R}). \quad (4)$$

(3) *Scattering: S.* A scattering process is defined to account for diffusion. A fraction r of the particles going in direction $\mathbf{e}_{\alpha}$ are scattered to directions $\mathbf{e}_{\alpha\pm 1}$, where $\alpha\pm 1$ is the velocity direction adjacent to direction α in the (+) clockwise, (-) counter clockwise direction [see Fig. 3(a)]. The scattering process occurs after translation, thus

$$\begin{aligned} F_{\alpha}(\mathbf{R}, t_{m+1}) = & [(1-2r)F_{\alpha}(\mathbf{R} + a\mathbf{e}_{\alpha+3}, t_m) \\ & + rF_{\alpha+1}(\mathbf{R} + a\mathbf{e}_{\alpha-2}, t_m) \\ & + rF_{\alpha-1}(\mathbf{R} + a\mathbf{e}_{\alpha+2}, t_m)] \phi(\mathbf{R}). \end{aligned} \quad (5)$$

(4) *Relaxation: R.* Magnetic relaxation at the boundaries of the pore space is defined using a transmission coefficient χ . Particles headed toward a site where $\phi(\mathbf{R})=0$ are transmitted and disappear (relax) with probability χ . They are reflected with probability $1-\chi$. The velocity of the reflected particles changes direction from $\mathbf{e}_{\alpha}$ to $\mathbf{e}_{\alpha+3}$ [see Fig. 3(b)]. The reflection process occurs after translation and scattering, thus

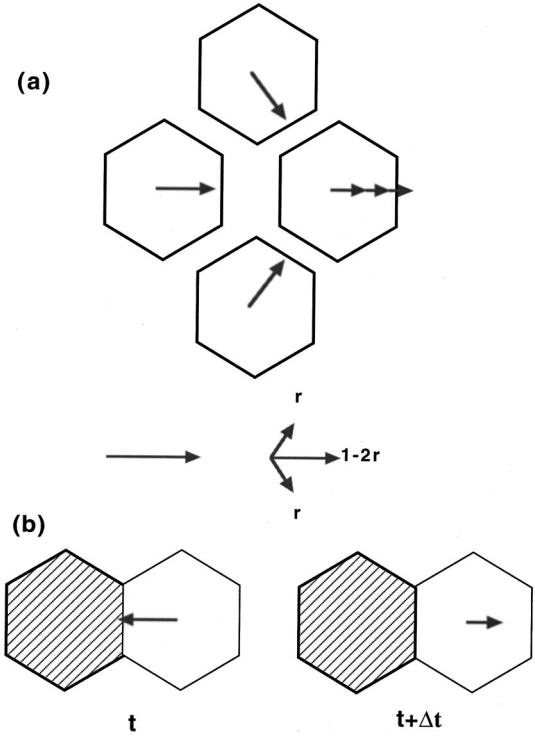


FIG. 3. Diffusive scattering and surface relaxation. (a) A scattering process that conserves energy but scatters momentum operates at each site. At each time step, fractions r of the particles moving in direction $\mathbf{e}_{\alpha}$ are scattered to directions $\mathbf{e}_{\alpha+1}$ and $\mathbf{e}_{\alpha-1}$; $r \leq 0.5$. The scattering process occurs after translation. The diffusion constant is related to the scattering rate, $D \propto 1/r$. (b) The relaxation of magnetization at the pore space-matrix interface is related to the transmission coefficient χ . In one time step, a fraction $1-\chi$ of the particles return to the pore space site going in the opposite direction. For $\chi=1$ all of the particles leave the pore space; for $\chi=0$ none of the particles leave the pore space.

$$\begin{aligned} F_{\alpha}(\mathbf{R}, t_{m+1}) = & \{ (1-2r)F_{\alpha}(\mathbf{R} + a\mathbf{e}_{\alpha+3}, t_m) \\ & + rF_{\alpha+1}(\mathbf{R} + a\mathbf{e}_{\alpha-2}, t_m) \\ & + rF_{\alpha-1}(\mathbf{R} + a\mathbf{e}_{\alpha+2}, t_m) \\ & + (1-\chi)F_{\alpha+3}(\mathbf{R}, t_m) \\ & \times [1 - \phi(\mathbf{R} + a\mathbf{e}_{\alpha+3})] \} \phi(\mathbf{R}). \end{aligned} \quad (6)$$

B. The hydrodynamic limit and calibration of r and χ

Our computational rules for the motion of magnetization, including scattering and relaxation, result in Eq. (6). To demonstrate that Eq. (6) describes diffusion and relaxation, we look at it in the hydrodynamic limit. Comparison of this limit to Eq. (1) establishes the relationships between the macroscopic parameters D and ρ , and the parameters r and χ employed in numerical work. To establish the proportionality constants for a particular computational realization of the time evolution, we perform computational experiments. A sketch of the steps in the derivation of the diffusion equation from the computational rule Eq. (5) is given in Appendix A. We find

$$D = \frac{a^2}{2\pi r} \quad (7)$$

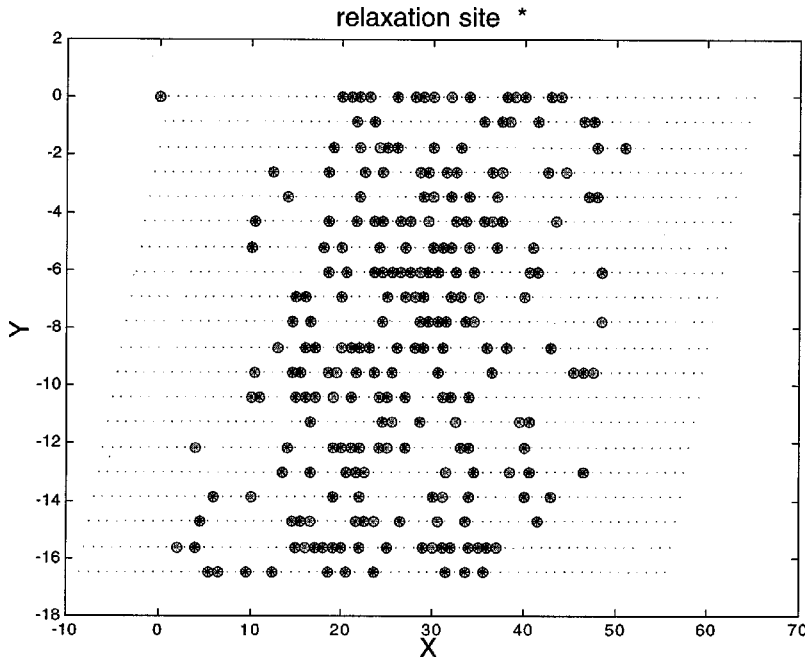


FIG. 4. Relaxation sites. Longitudinal magnetization is simulated for a periodic pore space having $20 \times 65 = 1300$ sites. The 231 filled circles are the matrix relaxation sites (except the isolated, filled circle at the upper left that locates $\mathbf{R}_{11}$). The region with no relaxation sites is taken to be a pore, the region with relaxation sites is called the microporous region. The relaxation sites in the microporous region are chosen from a random distribution with parabolic shape in the range $10 \leq l \leq 55$.

and

$$\rho = \frac{\chi a}{\tau}, \quad (8)$$

where ρ is the surface relaxivity, a velocity.⁵ These results are achieved employing approximations that extend beyond taking things to be slowly varying in space and time. Thus we use numerical experiments to establish the constants of proportionality in Eqs. (7) and (8).

To establish the relationship between the scattering rate r , and the diffusion constant D , we set up a steady particle flow and use the definition

$$\mathbf{j} = -D \nabla n, \quad (9)$$

where $\mathbf{j}$ is the particle flux, and n is the areal number density of particles. The system studied is $K \times L$ in size (L sites in the $\mathbf{e}_1$ direction, and K sites in the $\mathbf{e}_5$ direction), uniform [$\phi(\mathbf{R}) = 1$ for all sites], and periodic. A source of particles is placed at $L1 = 1$ [$F_\alpha(\mathbf{R}_{k,L1}, t) = 1.1$], and a sink of the same strength is placed at $L2 = 1 + L/2$ [$F_\alpha(\mathbf{R}_{k,L2}, t) = 0.9$]. The initial condition is $F_\alpha(\mathbf{R}, 0) = 1$. The source and sink are turned on at $t = 0$ and the time evolution of the distribution functions is followed until steady state is reached. Current flows in both directions from the source to the sink because the system is periodic. Thus there are two measureable steady currents in the $\mathbf{e}_1$ direction. The diffusion coefficient is calculated from the steady current configuration

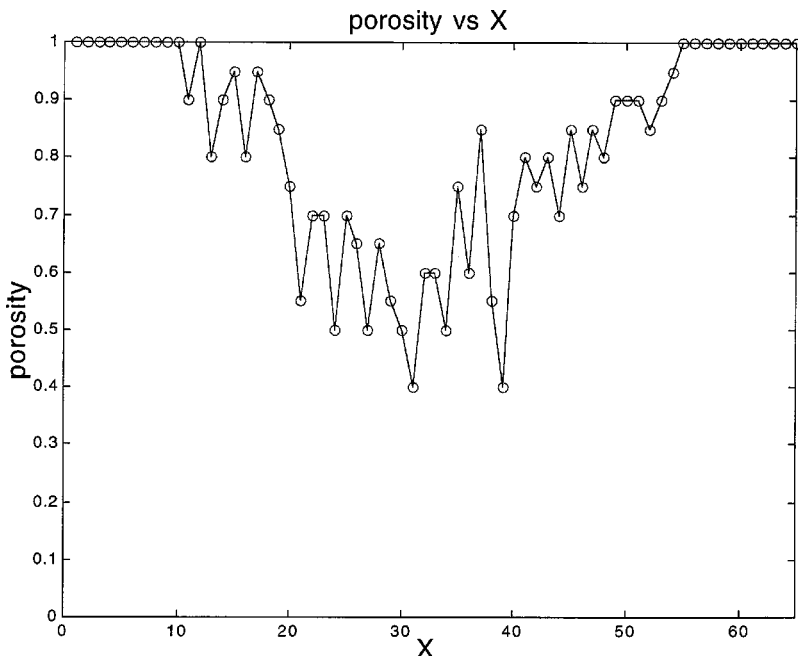


FIG. 5. Local porosity. The porosity of Fig. 4 averaged along lines of constant l . The magnetization, averaged along lines of constant l , looks similar to the porosity at early times. (See Fig. 8.)

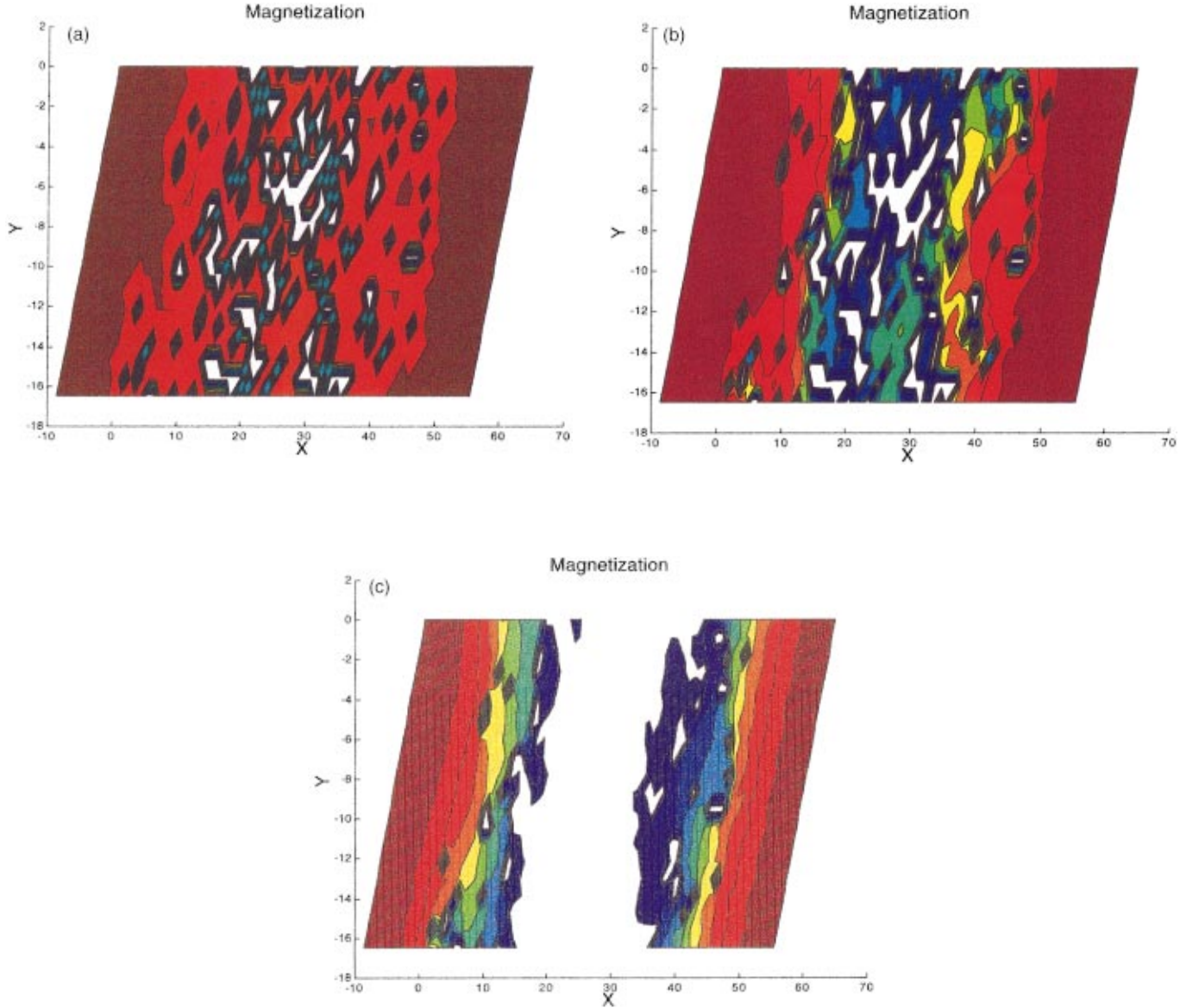


FIG. 6. (Color) Magnetization. The three panels show the magnetization of the system pictured in Fig. 4 at times $t=3\tau$, 30τ , and 300τ [panels (a), (b), and (c)]. Dark red corresponds to a magnetization per site approximately 1 at $t=3\tau$ and 30τ , and approximately 0.55 at $t=300\tau$. White corresponds to a magnetization per site less than 0.05.

$$D = \frac{\sqrt{3}ac \sum_{\alpha} F_{\alpha}(\mathbf{R}, t) \mathbf{e}_{\alpha} \cdot \mathbf{e}_1}{2 \sum_{\alpha} [F_{\alpha}(\mathbf{R} + a\mathbf{e}_{\alpha+3}, t) - F_{\alpha}(\mathbf{R} + a\mathbf{e}_{\alpha}, t)]}, \quad (10)$$

where a factor $\sqrt{3}/2$ comes from the cell area. The results of this numerical experiment show that D indeed scales with $1/r$ as in Eq. (7) with a numerical coefficient slightly less than $1/2$.

To establish the relationship between the transmission coefficient χ and the wall relaxation rate ρ , we use the boundary condition for surface relaxation⁵

$$(D\mathbf{e}_{\perp} \cdot \nabla n + \rho n)|_s = 0, \quad (11)$$

where $\mathbf{e}_{\perp}$ is the unit vector normal to the surface s . Using Eq. (9), we write

$$(-\mathbf{e}_{\perp} \cdot \mathbf{j} + \rho n)|_s = 0. \quad (12)$$

To use this relationship we employ a periodic system the same size as above. A source of particles is placed at $L1$ [$F_{\alpha}(\mathbf{R}_{k,L1}, t) = 1.1$], and a line of matrix sites crosses the system at $L2$ [$\phi(\mathbf{R}_{k,L2}) = 0$]. The initial condition is $F_{\alpha}(\mathbf{R}, 0) = \phi(\mathbf{R})$. The source is turned on at $t=0$ and the time evolution of the distribution functions is followed until steady state is reached. The wall relaxation rate is calculated in the steady state configuration

$$\rho = \frac{\sqrt{3}c \sum_{\alpha} F_{\alpha}(\mathbf{R}_s, t) \mathbf{e}_{\alpha} \cdot \mathbf{e}_{\perp}}{2 \sum_{\alpha} F_{\alpha}(\mathbf{R}_s, t)}, \quad (13)$$

where $\mathbf{R}_s$ are sites adjacent to the relaxation surface and a factor $\sqrt{3}/2$ comes from the cell area. The numerical rela-

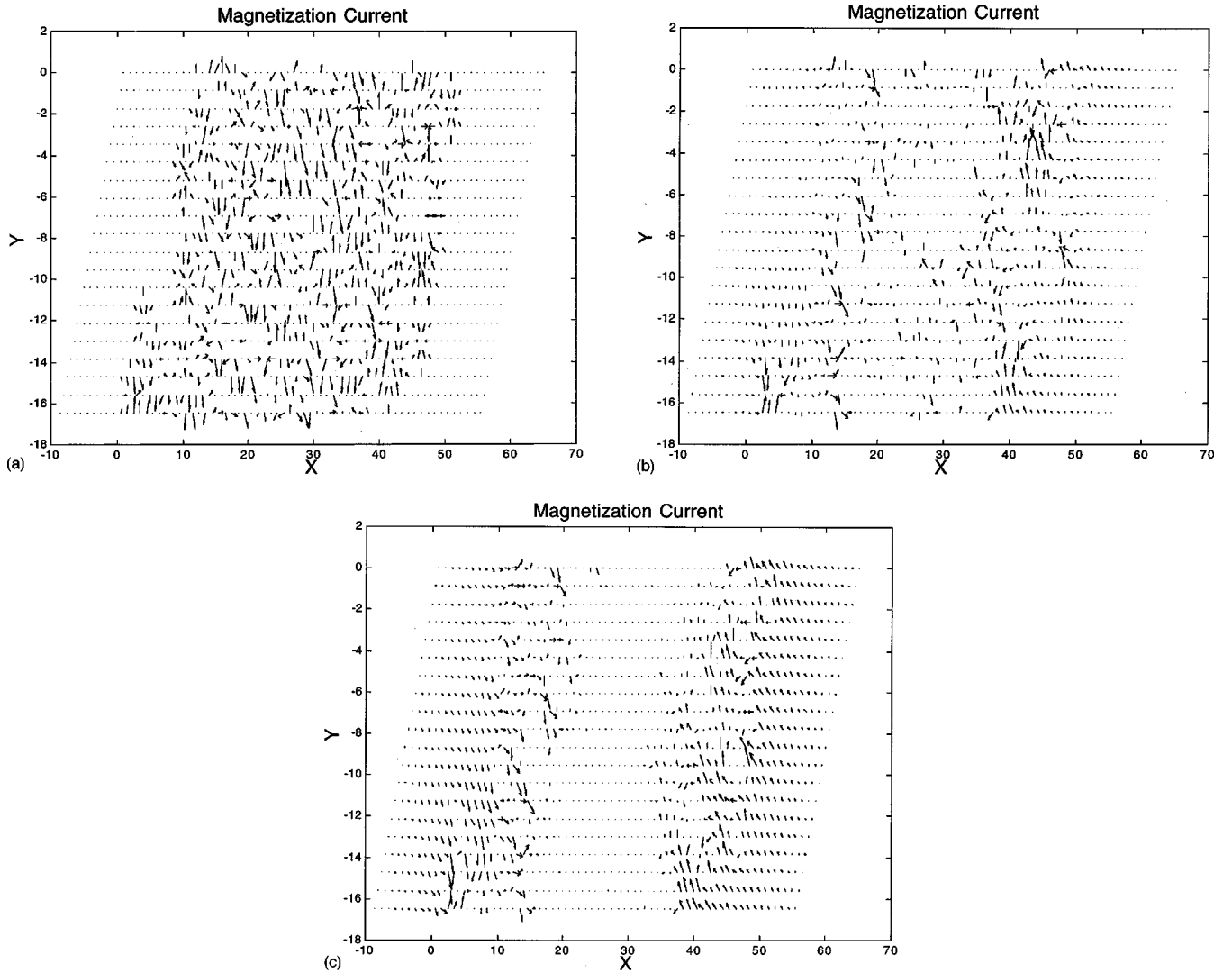


FIG. 7. Magnetization current. The three panels show the magnetization current at $t=3\tau$, 30τ , and 300τ [panels (a), (b), and (c)]. The magnitude of the current vectors is scaled so that the largest current flowing at each time corresponds to an arrow of unit length. By choosing the computational space as described in Fig. 4, the current flows on average perpendicular to lines of constant k , and appears in the figure to flow more in the y direction than in the x direction. Part of this appearance is due to the distortion of the x and y coordinates. Part of it is due to the choice of the computational space. The choice of the computational space was made in part so that the current vectors would not overlay one another.

tionship between χ and ρ is found to scale with χ , $\rho \approx \chi/3$, $0.01 \leq \chi \leq 1.0$. The factor $1/3$ comes from the precise geometry used in this test; each $\mathbf{R}_s$ site sees two matrix sites and delivers $2/6$ of its magnetization to the matrix. Thus, the numerical relationship between ρ and χ reminds us that the precise relationship depends on the details of the geometry of the matrix system, and should be numerically calibrated when careful numbers are at issue.

III. T_1 AND SCALAR MAGNETIZATION

A T_1 measurement involves (1) bringing the z magnetization into thermal equilibrium in an applied field [$M_z(t < 0) = M_\infty$], (2) applying a $\pi/2$ pulse at $t=0$ to set the z magnetization to zero [$M_z(0)=0$], and (3) measuring the recovery of the z magnetization to M_∞ . Numerically, we study $M(t) = M_\infty - M_z(t)$. This magnetization difference obeys Eq. (1), is M_∞ at $t=0$, and decays to zero as t advances. Thus our

initial condition is $F_\alpha(\mathbf{R}, 0) = \phi(\mathbf{R})$.

We study a simple model pore space, a pore adjacent to a microporous region, such as would be found in a carbonate rock.^{13,16} Matrix sites are randomly distributed in the middle of the $K \times L = 20 \times 65$ periodic space, as shown in Fig. 4. There are 231 matrix relaxation sites distributed between $l=11$ and $l=55$. The transmission and scattering parameters are $\chi=0.02$ and $r=0.5$ ($D \approx a^2/\tau$). Figure 4 also illustrates the geometry of the computational space that we use throughout this paper. The computational space is chosen to be a segment of space whose site variables are conveniently represented with the coordinates of a matrix. Recall, the notation is $\mathbf{R}_{kl} = (l-1)\mathbf{a} + (k-1)\mathbf{b}$, where $l=1, \dots, L$, $k=1, \dots, K$, $\mathbf{a} = a(1,0)$, $\mathbf{b} = a(-1/2, \sqrt{3}/2)$ in an x - y coordinate system, and a is the unit of length. The isolated point in the upper left is $\mathbf{R}_{11}$.

The time evolution of the magnetization was followed for 900 time steps (900τ), at which point the magnetization was

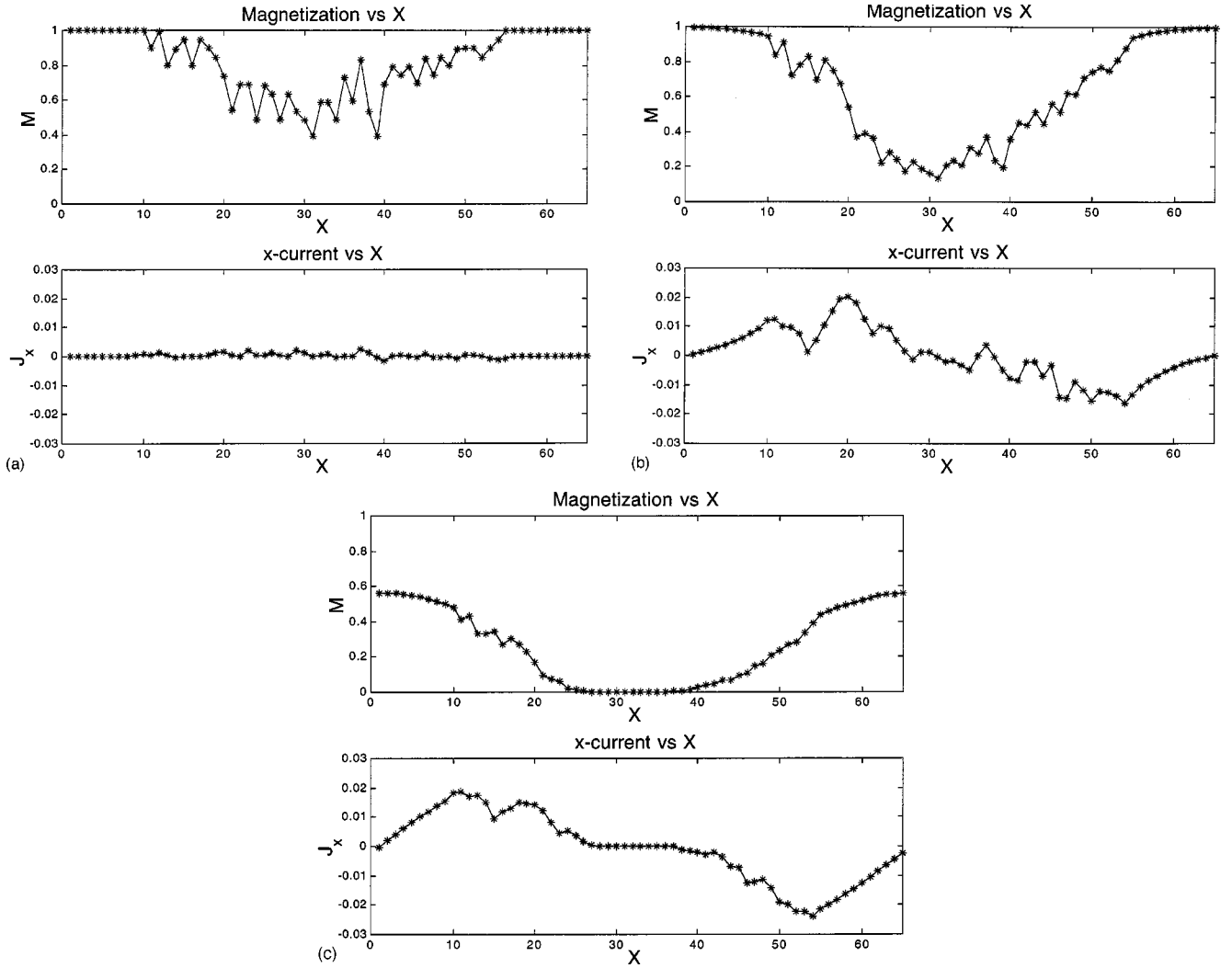


FIG. 8. Local magnetization and current. The three panels show the local magnetization and local magnetization current at $t=3\tau$, 30τ , and 300τ [panels (a), (b), and (c)] as a function of $x=l$. At $t=3\tau$ the magnetization is essentially the same as the porosity shown in Fig. 5, and the current is random, and thus very small. As time progresses a current builds up that feeds magnetization into the microporous region which starts at $l \approx 10$ and extends to $l \approx 55$. A penetration depth approximately 15 sites wide can be seen in both the current pattern and in the behavior of the magnetization.

below 3% of its initial value. In the following series of figures, we describe various measures of the behavior of the magnetization as the lattice Boltzmann calculation proceeds. The reported magnetization, either total or for a single site, is normed such that its equilibrium value is unity.

The local porosity of the system in Fig. 4 is defined by

$$\Phi(l) = \frac{1}{K} \sum_k \phi(\mathbf{R}_{kl}), \quad (14)$$

and shown in Fig. 5 as a function of $X=l$. The average porosity in the microporous region $11 < l < 55$ is 0.74, however, the local porosity fluctuates substantially, to values as low as 0.42. Outside the microporous region, the local and average porosity is unity.

In Figs. 6(a)–6(c) the magnetization $M(\mathbf{R}, t)$ at $t=3\tau$, 30τ , and 300τ is plotted as a contour plot with red (white) corresponding to the largest (smallest) magnetization values at each time. At the first two moments of time, $t=3\tau, 30\tau$, the magnetization in the microporous region is

decaying and coming into equilibrium with the flow of magnetization from the $\Phi=1$ regions of the pore space. The magnetization configuration in Fig. 6(c) is the asymptotic configuration; by $t=300\tau$ there is no magnetization in the heart of the microporous region, and the magnetization currents in the $\Phi=1$ regions of the pore space are toward the edges of the microporous region.

In Figs. 7(a)–7(c) the magnetization current $\mathbf{J}(\mathbf{R}, t)$ at $t=3\tau$, 30τ , and 300τ is plotted using arrows to indicate direction and magnitude. The magnetization current at the earliest moments of time is small and chaotic. By $t=30\tau$ the pattern of currents is becoming organized; there is much less current in the interior of the microporous region. As time advances to beyond $t=100\tau$ an organized current pattern that represents the long time steady flow of magnetization toward the microporous region is established.

The local magnetization and the local x component of the magnetization current are shown for $t=3\tau$, 30τ , and 300τ in Figs. 8(a)–8(c). These local properties are averages over k ,

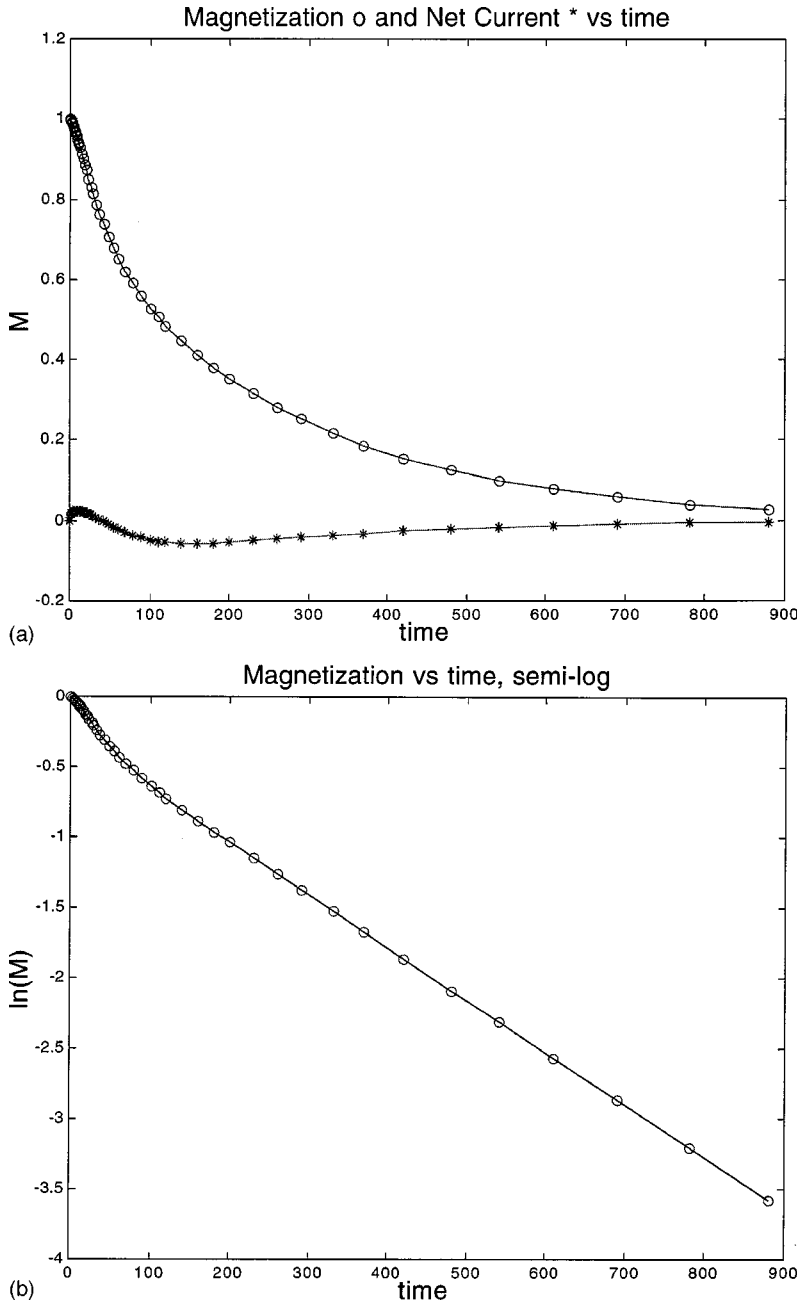


FIG. 9. Magnetization relaxation. (a) $M(t)$, the total magnetization in the system is plotted as a function of time, solid curve with open circles. The net current per site as a function of time is shown by the solid curve with stars. The net current is small and of varying sign as time evolves. (b) The magnetization as a function of time on a semilog scale. It is apparent that there are two relaxation times. The shorter is of order 100τ and the longer is of order 300τ . These relaxation times correspond to rapid relaxation of the magnetization in the microporous space and relatively slow relaxation of the magnetization initially outside of this space that must flow to it in order to relax.

$$M(l,t) = \frac{1}{6K} \sum_k \sum_\alpha F_\alpha(\mathbf{R}_{kl}, t),$$

$$J_x(l,t) = \frac{1}{6K} \sum_k \sum_\alpha F_\alpha(\mathbf{R}_{kl}, t) \mathbf{e}_\alpha \cdot \mathbf{e}_1, \quad (15)$$

and are plotted as a function of l ; the factor of 6 norms the equilibrium values to unity. At the earliest time, $t=3\tau$, the magnetization reflects the local porosity and the magnetization currents are small and random. At $t=30\tau$ the magnetization in the interior of the microporous region has decayed substantially while the magnetization at the edges of the pore space is still unaffected. There are magnetization currents toward the microporous region and within the microporous region. At $t=300\tau$ the magnetization in the heart of the microporous region is gone and the magnetization in the $\Phi=1$ region of the pore space has decayed to about 0.6.

Steady currents toward the microporous region and complex currents in a penetration depth within the microporous region are well established. The current patterns seen in Fig. 8(c) change in amplitude but not in structure as time evolves beyond $t=300\tau$.

In Fig. 9 we show the time evolution of the total magnetization,

$$M(t) = \frac{\sum_k \sum_l \sum_\alpha F_\alpha(\mathbf{R}_{kl}, t)}{\sum_k \sum_l \sum_\alpha F_\alpha(\mathbf{R}_{kl}, 0)}, \quad (16)$$

and the total x current. Figure 9(a) is a linear plot showing that the total x current is small, changes sign, and is asymptotically negative. Figure 9(b) is a semilog plot revealing that there are two characteristic times in the decay of the magne-

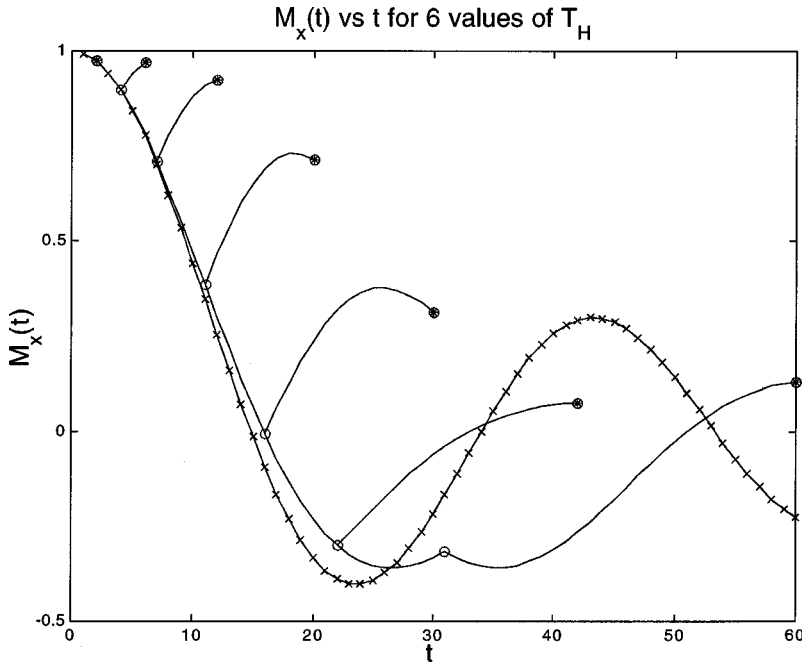


FIG. 10. Hahn echo magnetization. The six solid curves with open and closed circles are $M_x(t)$ for six choices of t_H ; t_H is marked by the open circle, and $2t_H$ by the filled circle on each curve. The solid curve with crosses is the analytic and numerical solution for the magnetization in the limit of stationary particles ($D=0$). At $t = 23$, $M_x(t) = J_0(\omega_0 t L / 2\pi)$ is a minimum. In this amount of time a particle has diffused a distance of order $\sqrt{2Dt} \approx 12a$.

tization. At early times, $t < 150\tau$, the magnetization decays on time scale $\tau_1 \approx 120\tau$; at late time, $t > 150\tau$, the magnetization decays on time scale $\tau_2 \approx 270\tau$. The short time scale is associated with relaxation in the microporous space. A simple estimate of the relaxation rate expected in this region yields

$$\frac{1}{\tau_1} \approx \rho \frac{S}{V} \approx \frac{\chi}{\tau} \frac{231}{449} \approx \frac{1}{100\tau}, \quad (17)$$

where S is the matrix surface area, and V is the pore volume in the microporous region. The long time scale is an effective relaxation time associated with relaxation at rate $1/\tau_1$ within a "volume" of the microporous region that is the product of the pore-micropore interface area and a penetration depth. The penetration depth L_P is controlled by D and τ_1 , $L_P = \sqrt{D\tau_1}$. We have studied the dependence of τ_2 on r and χ and confirmed

$$\frac{1}{\tau_2} \propto \frac{L_P}{\tau_1} \propto \sqrt{\frac{\chi}{r}}. \quad (18)$$

To study the decay of the magnetization we might write

$$M(t) = A_1 e^{-t/\tau_1} + A_2 e^{-t/\tau_2}, \quad (19)$$

However, it is clear from looking at the details of the behavior of $M(x, t)$ that the long time decay rate is associated with a configuration of the magnetization that is constructed as time evolves [it is not present at all times as Eq. (19) would imply]. A simple and perhaps adequate generalization of Eq. (19) makes A_2 a function of time

$$M(t) = A_1 e^{-t/\tau_1} + A_2(t) e^{-t/\tau_2}. \quad (20)$$

The physical picture implied by Eq. (20) and our discussion is due to Ramakrishnan *et al.*¹³

IV. T_2 AND VECTOR MAGNETIZATION

The transverse magnetization in the pore fluid has two components, M_x and M_y , that obey a coupled set of diffusion equations¹⁷

$$\begin{aligned} \frac{\partial M_x}{\partial t} &= D \nabla \cdot \phi \nabla \left(\frac{M_x}{\phi} \right) + \omega(\mathbf{x}) M_y, \\ \frac{\partial M_y}{\partial t} &= D \nabla \cdot \phi \nabla \left(\frac{M_y}{\phi} \right) - \omega(\mathbf{x}) M_x, \end{aligned} \quad (21)$$

where precession at $\mathbf{x}$, at frequency $\omega(\mathbf{x})$, is due to the z -magnetic field at $\mathbf{x}$. The precession at the Larmor frequency of the external field has been removed, and the bulk decay time T_{2B} is assumed long compared to other timescales in the problem and is neglected. The x and y components of the magnetization diffuse with the same diffusion constant through the same pore space. The local z -magnetic field converts x magnetization to y magnetization and y magnetization to x magnetization pointwise in space. The z field has three sources: relaxation sites on the pore walls; internal gradients; and applied gradients, where "gradient" is used to describe all local z -magnetic fields that depart from the average field determining the Larmor frequency.

(1) Relaxation sites. The transverse magnetization relaxation sites are local sources of substantial phase evolution on the pore space-matrix interface. They can be modeled in much the same way as longitudinal relaxation sites

$$\omega_R(\mathbf{x}) = \sum_i a_i^3 \omega_i \delta(\mathbf{x} - \mathbf{x}_i), \quad (22)$$

where ω_i is the strength of the precession induced at the relaxation site i . The resulting incoherent phase evolution degrades the transverse magnetization extremely rapidly locally. It is conventional to replace this process by a wall relaxation time⁵ $1/T_2^w$.

(2) Internal gradients. Internal gradients are due to the applied magnetic field's interaction with magnetic susceptibility contrasts within the matrix.¹⁸

(3) Applied gradients. Applied gradients are deliberately applied fields in the z direction.¹⁹ In the following, we consider only applied gradients.

To numerically model the transverse magnetization we adopt a two fluid model and most of the conventions established for studying longitudinal magnetization. One fluid carries the x component of the magnetization, and a second fluid carries the y component of the magnetization. The distribution functions $F_1, \dots, F_6$ describe the fluid carrying the x magnetization, and the distribution functions $G_1, \dots, G_6$ describe the fluid carrying the y magnetization. A local precession frequency $\omega(\mathbf{R}, t)$ is defined on each site $\mathbf{R}$ for which $\phi(\mathbf{R}) = 1$. Where $\omega(\mathbf{R}, t) = 0$, the two fluids are uncoupled; they evolve in time according to the translation and scattering rules developed for longitudinal magnetization, and there is no net change in either fluid. Where $\omega(\mathbf{R}, t) \neq 0$, there is a conversion (C) of one fluid into the other,

$$\begin{aligned} F_\alpha(\mathbf{R}, t_{m+1}) &= \cos \theta(\mathbf{R}, t_m) F_\alpha(\mathbf{R}, t_m) \\ &\quad + \sin \theta(\mathbf{R}, t_m) G_\alpha(\mathbf{R}, t_m), \\ G_\alpha(\mathbf{R}, t_{m+1}) &= \cos \theta(\mathbf{R}, t_m) G_\alpha(\mathbf{R}, t_m) \\ &\quad - \sin \theta(\mathbf{R}, t_m) F_\alpha(\mathbf{R}, t_m), \end{aligned} \quad (23)$$

where $\theta(\mathbf{R}, t) = \tau \omega(\mathbf{R}, t)$. The time evolution of the distribution functions has the basic time step of translation, scattering, and conversion. A sequence of these steps produces the complete time evolution. Manipulations of NMR fields are manipulations of the conversion process. Below we describe the application of the lattice Boltzmann computational rules for transverse magnetization to two examples: (A) a Hahn echo scenario and (B) a pulsed field gradient scenario.¹⁹

A. Simulation of a Hahn echo experiment

To study the time evolution of the transverse magnetization we consider a $K \times L$ space (where L is odd) in a periodic magnetic field,

$$\omega(\mathbf{R}_{kl}) = \frac{\omega_0 L}{2\pi} \sin \frac{2\pi(l - L_{\text{mid}})}{L}, \quad (24)$$

where $L_{\text{mid}} = (L + 1)/2$. For $|l - L_{\text{mid}}| \ll L$, $\omega = \omega_0(l - L_{\text{mid}})$, i.e., there is a linear gradient near the center of the system. The pore space is uniform, $\phi(\mathbf{R}) = 1$, and periodic.

In the Hahn echo simulation, the transverse magnetization is in equilibrium with $M_x = 0$ and $M_y = 0$ for $t < 0$. A $\pi/2$ pulse at $t = 0$ creates x magnetization, i.e., $M_x = M_\infty$ and $M_y = 0$. Thus, in terms of the distribution functions, the initial conditions are $G_\alpha(\mathbf{R}, 0) = 0$ and $F_\alpha(\mathbf{R}, 0) = 1$. As time evolves the distribution functions evolve by translation, scattering, and conversion, and the x magnetization M_x initially decays. The operator CST will be used to describe this evolution. At $t = t_H$ the magnetization is subjected to a π pulse; this is accomplished numerically by flipping $\omega(\mathbf{R})$ to $-\omega(\mathbf{R})$ at $t = t_H$. Thus for $t > t_H$, the basic time step is $C^{-1}ST$, where C^{-1} denotes conversion according to Eq. (23) with $\theta(\mathbf{R}, t > t_H) = -\tau \omega(\mathbf{R})$. At $t = 2t_H$, M_x reaches a

local maximum called the Hahn echo. A schematic equation for the Hahn echo amplitude is

$$M_x(2t_H) = [C^{-1}ST]^{N_H} [CST]^{N_H} M_x(0), \quad (25)$$

where $N_H = t_H / \tau$.

The evolution of x magnetization is shown for six values of t_H in Fig. 10. The numerical parameters are $K = 21$, $L = 51$, $r = 0.15$, $\chi = 0$, and $\omega_0 \tau = 0.02$. Also shown are the analytic and numerical solutions for $M_x(t)$ for stationary particles ($D = 0$). The derivation of the analytic solution $M_x(t) = J_0(\omega_0 t L / 2\pi)$ is sketched in Appendix B. The numerical solution is found by eliminating translation and scattering from the timesteps and allowing only conversion. In Fig. 10, the $D \neq 0$ results show modest decay away from the $D = 0$ curve.

A measurement of the time evolution of the transverse magnetization consists of recording the amplitude of the Hahn echo, $M_x(t = 2t_H)$, for a sequence of values of t_H . In

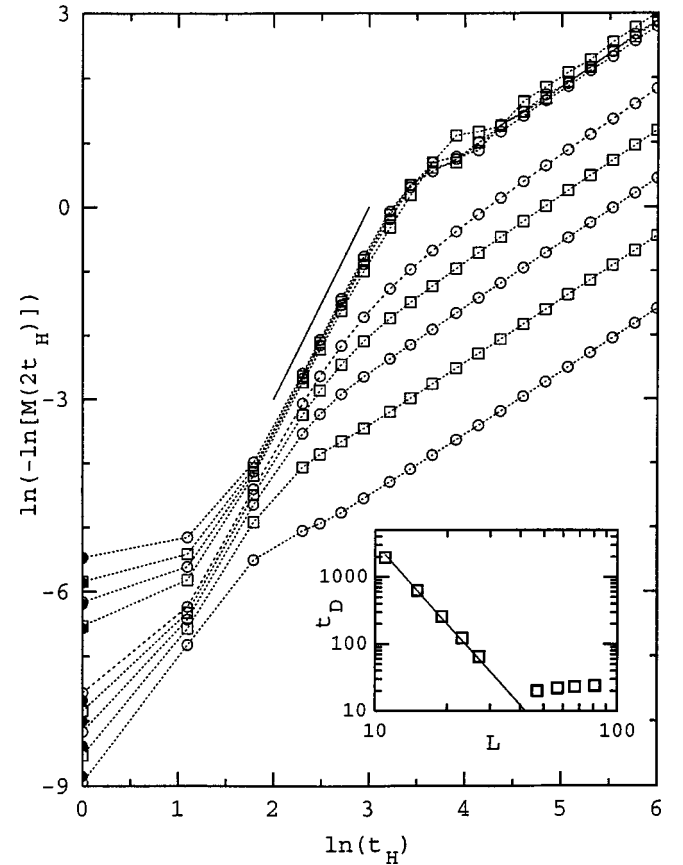


FIG. 11. Hahn echo amplitude versus time and system size. The echo amplitude $\ln\{-\ln[M_x(2t_H)]\}$ as a function of $\ln(t_H)$ for nine system sizes ($L = 11, 15, 19, 23, 27, 47, 57, 67, 81$). The lowest curve corresponds to the smallest system size. The five independent curves are for the five smallest system sizes, $L < 40$. The four curves that overlay one another at intermediate and long time are for the largest system sizes, $L > 40$. The universal behavior for large system size corresponds to the localization limit. The solid line corresponds to the expected unrestricted diffusion limit, $M \propto \exp(at^3)$. The solid circles at $\ln(t_H) = 0$ are the stationary particle results for short time (Appendix B). The inset shows τ_D from a fit to the asymptotic behavior of the magnetization, $M = \exp(t/\tau_D)$. The solid line in the inset is L^{-4} .

Fig. 11 we show $\ln\{-\ln[M_x(2t_H)]\}$ versus $\ln(t_H)$ for nine values of L between 11 and 81. The other numerical parameters are $K=21$, $r=0.25$, $\chi=0$, and $\omega_0\tau=0.01$. The results are reported as log-log versus log so that the time dependence of the decay of the echo amplitude is emphasized. Figure 11 shows evidence for a crossover, as a function of system size, from a time scale that is system size dependent to a time scale that is universal.^{7,20}

(1) At very short times, i.e., $t_H = \tau$, the magnetization evolution will be dominated by the conversion process rather than diffusion. Thus we expect the stationary particle result of Appendix B to be a good estimate of the early time behavior in Fig. 11. For $\ln t_H = \ln \tau = 0$, the stationary particle result is $\ln[-\ln(M)] \rightarrow 2 \ln L - 14.27$. The filled circles on Fig. 11 show this result for the system sizes in question.

(2) For long times, the echo height decay is exponential with time, $M(2t_H) = \exp(-t_H/\tau_D)$. The inset in Fig. 11 shows the decay time τ_D as a function of system size L . The five smallest systems relax as $\tau_D \propto 1/L^4$, a characteristic dependence associated with diffusion in a bounded medium.²¹ For the four largest systems, τ_D is independent of system size. In these cases the system is said to be in the localization regime^{7,20} in which the dephasing length is shorter than the diffusion length and the system size.

(3) At intermediate times, the magnetization decays as t_H^3 , the unrestricted diffusion result for a Hahn echo.¹⁷ The region where $\ln[-\ln M(2t_H)] \sim 3 \ln(t_H)$ is longest for the large system sizes.

(4) The crossover from intermediate time dependence to asymptotic time dependence should be system size dependent for the small systems, with crossover time $t_c \propto L^2/D$. This is qualitatively true, but not quantitatively testable given the short time over which unrestricted diffusion occurs in these systems. The crossover time for the large systems occurs at a universal time $t_c \approx e^{-1}$. Studies of the dependence of the crossover time on r and ω_0 confirm the theoretical description of de Swiet and Sen⁷ (see Appendix B).

B. Simulation of a pulsed field gradient experiment

An alternative manipulation of the NMR fields for study of the time evolution of the transverse magnetization is the pulsed field gradient (PFG) pulse sequence. A schematic equation for the PFG echo amplitude is

$$M_x(t_E) = C^{-1}[ST]^{N_E}CM_x(0). \quad (26)$$

An applied field is turned on for one time step at $t=0$, allowing conversion (C). The field is turned off for N_E time steps, during which diffusion takes place (ST). Finally, the field is turned on again in the opposite direction for one time step (C^{-1}). The x magnetization is measured at $t=(N_E+2)\tau = t_E + 2\tau$.

To numerically simulate the PFG pulse sequence, we use a periodic system with a matrix boundary at $l=1$ [$\phi(\mathbf{R}_{k,1})=1$]. The system can be thought of as a one-dimensional slab of bounded fluid. We set $\chi=0$, thus there is perfect reflection of the particles approaching the matrix from both directions at $l=1$. A linear gradient

$$\omega(\mathbf{R}_k l) = \omega_0(l - L_{\text{mid}}), \quad (27)$$

is applied. In Fig. 12 we show $M_x(t_E)$ for several system sizes. The top curve is for the smallest system. In very little time the magnetization ceases to decay for this system. This behavior is characteristic of bounded diffusion.²¹ For larger system sizes, the behavior is essentially the same. However, there is no apparent systematic behavior to the asymptotic value of $\ln M_x(t_E)$.

To understand the asymptotic behavior, consider the Brownstein-Tarr description of magnetization evolution in an isolated pore (Appendix B). This description yields

$$M_x(t_E) \propto \left(\frac{\sin(\tau\omega_0 L/2)}{(\tau\omega_0 L/2)} \right)^2 \quad (28)$$

for long times. In Fig. 13 we show the asymptotic value of $M_x(t_E)$ as a function of L from the numerical study and the

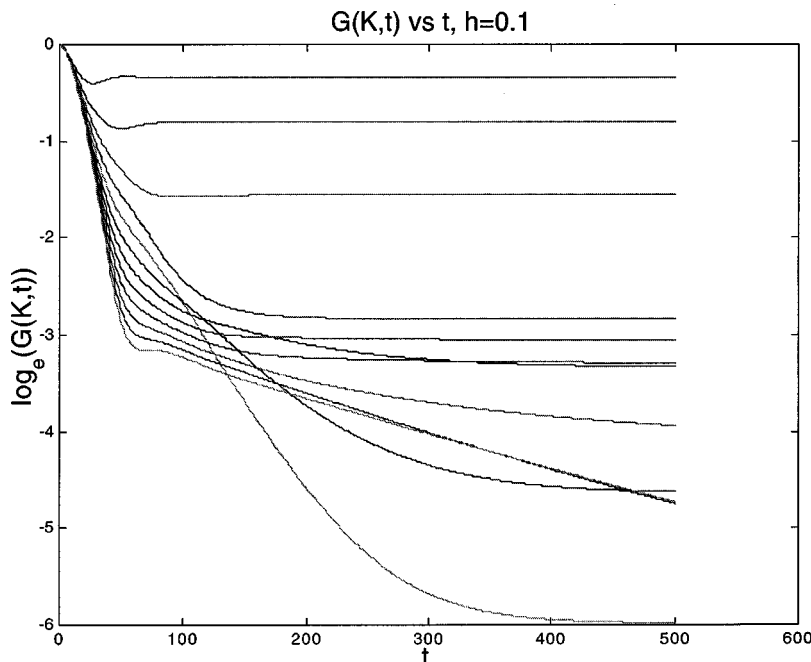


FIG. 12. Pulsed field gradient amplitude. The value of $\ln[M_x(t_E)]$ as a function of t_E for various system sizes between $L=11$ and $L=131$. The curve showing the least decay at long times corresponds to $L=11$. The magnetization curves depart from an inverted parabola systematically at larger time with increasing system size. However, the large time behavior is not superficially systematic due to diffraction (see Fig. 13).

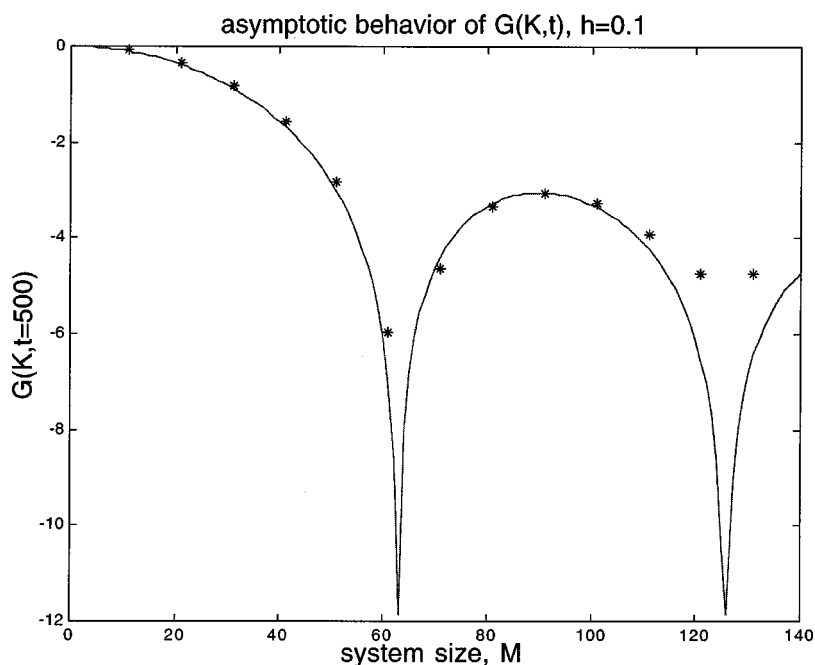


FIG. 13. Asymptotic behavior of PFG amplitude. The stars are the magnetization amplitude M_x at $t_E = 500$ as a function of system size L . The curve is the Brownstein-Tarr result of Eq. (28).

analytic result in Eq. (28). The agreement is excellent. The sort of chaotic behavior seen in Fig. 12 is a consequence of the relationship between the wave vector of the PFG exploration and the geometry of the pore. Figure 13 in conjunction with Eq. (28) illustrates how this kind of measurement can lead to information about the pores space geometry.

V. CONCLUSION

We have developed a lattice Boltzmann procedure to numerically simulate the time evolution of magnetization in a porous medium. The essential ingredients of the simulations are (1) a complex pore space in which the magnetization resides, (2) diffusive exploration of the pore space by the particles carrying the magnetization, (3) relaxation of the magnetization at the pore space-matrix interface, and (4) manipulation of the magnetization by various NMR fields.

Our computational procedure employs a set of rules for evolving the distribution functions appropriate to a coarse grained picture of the particle system. Time is coarse grained on scale τ , the pore space-matrix system (porous medium) is coarse grained on scale a , and velocity space is coarse grained by contraction to six velocity vectors of equal magnitude (in two dimensions).

The rules for the evolution of the distribution functions in one time step involve translation in discrete steps of amplitude a , a scattering process of strength r to account for particle diffusion, and a transmission coefficient χ , operating at the pore space-matrix interface, to account for surface relaxation. These computational rules can be reduced to hydrodynamic equations in the $k \rightarrow 0$ and $\omega \rightarrow 0$ limits, e.g., a continuum diffusion equation results from the Boltzmann limit. The relationship between parameters in the computational scheme, and parameters in the hydrodynamic equations is established by conducting computational experiments.

The lattice Boltzmann procedure can be used to study the time evolution of longitudinal magnetization in an arbitrarily complex porous material. In this case, a single fluid diffuses in the pore space and relaxes (disappears) due to interactions

at the pore walls. When the particles in the system carry an internal degree of freedom, as they do in the case of transverse magnetization, a two fluid model is employed. Each fluid carries one of the components of the internal degree of freedom. In the case of transverse magnetization one fluid carries the x magnetization (x fluid) and a second fluid (y fluid) carries the y magnetization. Field gradients and rf fields are coupled conveniently to the internal degree of freedom in a lattice Boltzmann calculation. The precession of transverse magnetization maps onto a conversion process between the x fluid and the y fluid that occurs pointwise throughout the pore space. Consequently the influence of complicated internal and external precession fields on the time evolution of the transverse magnetization can be described. Lattice Boltzmann methods are also well suited to problems in which a flow field is present.²²

A real experiment on the time evolution of magnetization is a very powerful tool. However, the results are necessarily an average over mesoscopic to macroscopic length scales. Much more detail can be probed in a computational experiment. The real experiment is telling you about the behavior of fluid magnetization in a real material, e.g., a Berea sandstone. The computational experiment is telling you about the behavior of fluid magnetization in a model material. The point of studying the model material is to look in great detail at the magnetization evolution, and to glean an understanding from what you see. This understanding may lead to better models of the real experiments, suggest useful manipulations of the NMR fields, suggest methods of data analysis, etc. Thus, numerical studies of model materials complement experimental studies of real materials. The lattice Boltzmann computational procedure is well suited to this task.

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APPENDIX A: THE HYDRODYNAMIC LIMIT

An example of the basic equation for the motion of the distribution function is

$$F_\alpha(\mathbf{R}, t_{m+1}) = [(1-2r)F_\alpha(\mathbf{R} - a\mathbf{e}_\alpha, t_m) + rF_{\alpha+1}(\mathbf{R} - a\mathbf{e}_{\alpha+1}, t_m) + rF_{\alpha-1}(\mathbf{R} - a\mathbf{e}_{\alpha-1}, t_m)], \quad (\text{A1})$$

where the distribution functions translate and then scatter. We write this in the form

$$e^{\tau L} F_\alpha(\mathbf{R} - a\mathbf{e}_\alpha, t_m) = \sum_{\beta} T(\mathbf{RS})_{\alpha\beta} F_\beta(\mathbf{S}, t_m), \quad (\text{A2})$$

where

$$\tau L = \tau \frac{\partial}{\partial t} + a\mathbf{e}_\alpha \cdot \nabla \quad (\text{A3})$$

is the Boltzmann translation operator and

$$S_n = \begin{pmatrix} 0.4082 & 0.5000 & 0.2887 & 0.5000 & -0.2887 & 0.4082 \\ 0.4082 & 0.5000 & -0.2887 & -0.5000 & -0.2887 & -0.4082 \\ 0.4082 & 0.0000 & -0.5774 & 0.0000 & 0.5774 & 0.4082 \\ 0.4082 & -0.5000 & -0.2887 & 0.5000 & -0.2887 & -0.4082 \\ 0.4082 & -0.5000 & 0.2887 & -0.5000 & -0.2887 & 0.4082 \\ 0.4082 & 0.0000 & 0.5774 & 0.0000 & 0.5774 & -0.4082 \end{pmatrix}, \quad (\text{A8})$$

have eigenvalues $\lambda_\nu/r = 0, 1, 1, 3, 3, 4$ from left to right, and are denoted as $\psi_0, \dots, \psi_5$ from left to right. The ground state, $\psi_0(\alpha)$ with $\lambda_0 = 0$, is the uniform state; it is not degraded by the collision process.

We develop $F_\alpha(\mathbf{R}, t)$ in terms of the set of vectors that describe the collision process

$$F_\alpha(\mathbf{R}, t) = \sum_\nu c_\nu(\mathbf{R}, t) \psi_\nu(\alpha). \quad (\text{A9})$$

In writing this equation we are describing the velocity structure of the distribution function with the ψ_ν ; the space and time structure appears in the coefficients $c_\nu(\mathbf{R}, t)$. Using Eq. (A9) in Eq. (A5) and the algebra appropriate to a complete set of orthonormal vectors leads to

$$\tau \frac{\partial c_\nu}{\partial t} + a \sum_\mu \sum_\gamma E_{\nu\mu}^\gamma \frac{\partial c_\mu}{\partial R_\gamma} = -\lambda_\nu c_\nu, \quad (\text{A10})$$

where

$$T(\mathbf{RS})_{\alpha\beta} = (1-2r)\delta_{\beta,\alpha}\delta_{\mathbf{S},\mathbf{R}-a\mathbf{e}_\alpha} + r[\delta_{\beta,\alpha-1}\delta_{\mathbf{S},\mathbf{R}-a\mathbf{e}_{\alpha-1}} + \delta_{\beta,\alpha+1}\delta_{\mathbf{S},\mathbf{R}-a\mathbf{e}_{\alpha+1}}] \quad (\text{A4})$$

is the collision operator.

The hydrodynamic limit means slowly varying in space and time. Thus τ and a are small; we use $\exp(\tau L) = 1 + \tau L$ on the left-hand side of Eq. (A2) and $\mathbf{S} = \mathbf{R}$ in the collision operator to write

$$\tau \frac{\partial F_\alpha(\mathbf{R}, t)}{\partial t} + a\mathbf{e}_\alpha \cdot \nabla F_\alpha(\mathbf{R}, t) = \sum_\beta T_{\alpha\beta} F_\beta(\mathbf{R}, t), \quad (\text{A5})$$

where

$$T_{\alpha\beta} = -2r\delta_{\beta,\alpha} + r\delta_{\beta,\alpha-1} + r\delta_{\beta,\alpha+1}. \quad (\text{A6})$$

Equation (A5) is solved using the eigenstates of the collision operator. These are the six component vectors $\psi_\nu(\alpha)$ ($\alpha = 1, \dots, 6$) that solve

$$\sum_\beta T_{\alpha\beta} \psi_\nu(\beta) = -\lambda_\nu \psi_\nu(\alpha); \quad (\text{A7})$$

i.e., the six vectors that can be constructed from $\exp(i\nu\alpha)$ with ν such that $\nu(\alpha+6) = \nu\alpha + 2\pi n$. The vectors ψ_ν are the columns of the matrix

$$E_{\nu\mu}^\gamma = \sum_\alpha \psi_\nu(\alpha) e_\alpha^\gamma \psi_\mu(\alpha), \quad (\text{A11})$$

and $\nu, \mu = 0, \dots, 5$. The most important of the equations in Eq. (A10) is the equation for c_0 which does not decay due to the collision process. We write the system of equations in the form

$$\tau \frac{\partial c_0}{\partial t} + a \sum_\mu E_{0\mu}^x \frac{\partial c_\mu}{\partial R_x} = 0 \quad (\text{A12})$$

and

$$a E_{\nu 0}^x \frac{\partial c_0}{\partial R_x} = -\lambda_\nu c_\nu. \quad (\text{A13})$$

In writing these equations we have assumed the time dependence to be slow compared to the time scale for decay of the nonuniform modes ($\tau \partial c_\nu / \partial t \ll \lambda_\nu$), taken the coupling to the ground state to be most important (we have ignored $E_{\nu\mu}$

when neither ν or μ is the ground state), and chosen to consider a single direction in space, x . Combining Eqs. (A12) and (A13) we have

$$\frac{\partial c_0}{\partial t} - \frac{a^2}{\tau} \left(\sum_{\nu} \frac{E_{0\nu}^x E_{\nu 0}^x}{\lambda_{\nu}} \right) \frac{\partial^2 c_0}{\partial R_x^2} = 0. \quad (\text{A14})$$

The matrix elements necessary to determine the diffusion constant can be calculated numerically or analytically with the result

$$\sum_{\nu} \frac{E_{0\nu}^x E_{\nu 0}^x}{\lambda_{\nu}} = \frac{1}{2r} \quad (\text{A15})$$

or

$$D = \frac{a^2}{2\pi r}. \quad (\text{A16})$$

APPENDIX B: SPECIAL CASES OF MAGNETIZATION EVOLUTION

1. Stationary particles in a periodic field

Consider the transverse magnetization evolution of stationary particles in a z magnetic field. The diffusion coefficient $D=0$, and Eq. (21) reduces to coupled harmonic equations. The x magnetization is given by

$$\frac{\partial^2 M_x}{\partial t^2} = -\omega^2(\mathbf{x}) M_x. \quad (\text{B1})$$

The periodic field in the Hahn echo example can be written

$$\omega(x) = \frac{\omega_0 L}{2\pi} \sin \frac{2\pi(x - La/2)}{La}, \quad (\text{B2})$$

where La is the length of the system in the x direction. The phase evolution at x is $\cos(\omega t)$, and the total normed x magnetization in the system at time t is

$$M_x(t) = \frac{1}{La} \int_0^{La} dx \cos(\omega t) = \frac{1}{\pi} \int_0^{\pi} d\theta \cos(z \sin \theta) = J_0(z), \quad (\text{B3})$$

where $z = \omega_0 t L / 2\pi$. In the limit $z \rightarrow 0$, $M_x(t) \rightarrow 1 - z^2/4$, and $\ln\{-\ln[M_x(t)]\} \rightarrow 2 \ln(z/2)$.

2. Crossover to asymptotic time dependence

The Hahn echo magnetization evolution of large systems in a linear gradient has a crossover time from behavior characteristic of unrestricted diffusion, to localization behavior.

This crossover time is dependent on the diffusion coefficient of the particles, and the magnetization gradient strength²⁰

$$t_c \propto \frac{1}{D^{1/3}(\gamma G)^{2/3}}. \quad (\text{B4})$$

The correspondence between our numerical parameters and the analytic parameters is $D \propto 1/r$, and $\gamma G \propto \omega_0$. Thus, we expect

$$t_c \propto \frac{r^{1/3}}{h^{2/3}}. \quad (\text{B5})$$

The results of numerical calculations, such as that shown in Fig. 11, confirm the dependence in Eq. (B5).

3. PFG in an isolated pore

A pulsed field gradient pulse sequence using a linear gradient results in transverse magnetization $M_x(t_E) \propto G(\kappa, t_E)$ where G is the intermediate scattering function,¹⁰

$$G(\kappa, t) = \langle e^{i\kappa\alpha(t)} e^{-i\kappa\alpha(0)} \rangle, \quad (\text{B6})$$

$\alpha(t)$ is a particle position, and the wavevector κ is related to the conversion process described by Eq. (23) through $\theta(\mathbf{R}) = \tau\omega \equiv \kappa(x - L/2)$.

In the Brownstein-Tarr formalism^{5,10} for magnetization in an isolated one-dimensional pore with no decay at the pore boundaries, the equation for the intermediate scattering function is

$$G(\kappa, t) = \sum_{\nu} |\langle 0 | e^{i\kappa x} | \psi_{\nu} \rangle|^2 e^{-r_{\nu} t}, \quad (\text{B7})$$

where $|0\rangle$ is the initial state of uniform magnetization and the states $|\psi_{\nu}\rangle$ are the solutions to

$$D \frac{\partial^2 \psi_{\nu}}{\partial x^2} = -r_{\nu} \psi_{\nu}, \quad (\text{B8})$$

with no-flow boundary conditions $\partial \psi_{\nu} / \partial x|_{\pm L/2} = 0$. Thus, $\psi_{\nu}(x) \propto \cos[\nu\pi(x + L/2)/L]$, and the decay rate for each state is $r_{\nu} = D(\nu\pi/L)^2$. Asymptotically in time, only the ground state $\nu=0$ contributes to the intermediate scattering function. Thus for long times

$$G(\kappa, t) = \langle 0 | e^{i\kappa x} | \psi_0 \rangle \langle \psi_0 | e^{-i\kappa x} | 0 \rangle = \left(\frac{\sin(\kappa L/2)}{\kappa L/2} \right)^2. \quad (\text{B9})$$

This description employs the normal modes of the magnetization. One can view the PFG experimental procedure as a probe of matrix elements of these normal modes.^{10,11}

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